UMKC Impact & Translation Series

Funding to Move Research Toward Real-World Impact

Helping you move your work beyond the university

Research with real-world potential often needs resources to move forward. Whether that next step involves technical validation, customer discovery, prototype development, regulatory planning, partnership building, or startup formation, progress usually depends on securing the right support at the right time.

In the UMKC Impact & Translation Series, funding should be understood as a tool for reducing uncertainty and enabling progress. Funding is not the goal by itself. It is one of the mechanisms that can help move research, innovation, and creative work beyond the university and closer to real-world use.

For some projects, that support may come through grants, proof-of-concept funds, or university-backed programs. For others, it may involve angel investment, venture capital, strategic partnerships, or revenue-based approaches. The right source depends on the nature of the work, the pathway to impact, and the milestone that needs to be reached next.

UITS Whitepaper

This whitepaper is part of the **UMKC Impact & Translation Series (UITS)** - a set of short videos and supporting materials designed to help faculty think more clearly about how their work can move beyond the university and into real-world use.

The videos introduce core ideas. This document goes deeper on one of those ideas: **how different types of funding support the next stage of moving your work toward real-world use.**

This whitepaper outlines the major categories of funding that may support translation, commercialization, and startup development at UMKC. It is designed to help faculty and graduate students think more clearly about which funding sources fit which stage of progress, what those sources are intended to support, and when it makes sense to engage the Office of Commercialization in exploring next steps.

Grant-Based Funding

Grant funding is often the most appropriate early source of support for university-connected innovations because it can help reduce risk without requiring founders to give up ownership. In many cases, grants support the work that must happen before a startup, licensing effort, or external partnership is viable. That may include feasibility studies, prototype development, customer discovery, validation work, or early commercialization planning.

Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) Grants

The SBIR and STTR programs offer federal funding to small businesses conducting research and development with the potential for commercialization. These grants are highly competitive but provide significant financial support:

- **Phase I Grants**: Typically up to $150,000, supporting feasibility studies.
- **Phase II Grants**: Can reach up to $1 million, funding further development.
- **Phase III**: While no direct funding is provided, businesses that reach this stage often secure contracts with government agencies.

SBIR/STTR grants provide substantial funding without requiring dilution of equity, allowing university-affiliated entrepreneurs to explore market applications of their research.

Additionally, STTR grants require collaboration with a nonprofit research institution, making them ideal for university researchers who want to work with industry partners while maintaining strong ties to their academic work.

Successful SBIR/STTR applications demonstrate:

- A clear path to commercialization
- Strong market validation
- Feasible technical milestones and a well-structured team
- Potential customer engagement
- Letters of support from industry partners or potential customers

Many faculty researchers seek SBIR/STTR grants to fund technology validation, prototype development, and initial market testing, which are necessary steps before securing private investment or forming a startup.

Government Granting Agencies and the Current Funding Environment

Federal agencies such as the National Institutes of Health (NIH), National Science Foundation (NSF), Department of Energy (DOE), and Department of Defense (DOD) provide substantial grant funding for early-stage research and commercialization. These agencies support university-affiliated startups, offering both early-stage development grants and later-stage commercialization grants.

However, the landscape of government grant funding is currently in flux. With ongoing shifts in administration policies, research priorities, and budget allocations, funding availability and focus areas may change. Startups should closely monitor federal funding announcements, participate in grant workshops, and leverage university resources to stay ahead of policy changes that might impact their eligibility for funding.

Over the last few years, federal funding priorities have shifted to focus on strategic areas that present strong commercialization opportunities. Some notable trends include:

- **Artificial Intelligence (AI) and Machine Learning**: Many federal agencies, including NSF and NIH, are allocating significant funding for AI applications in healthcare, security, and manufacturing.
- **Climate Tech and Sustainability**: DOE and NSF have increased funding for energy storage, renewable energy solutions, and decarbonization technologies.
- **Biotechnology and Precision Medicine**: NIH and DoD are investing in personalized medicine, genomic research, and bioinformatics to advance targeted healthcare solutions.
- **Supply Chain Resilience**: Federal funding programs now emphasize advanced manufacturing and logistics solutions to address global supply chain challenges.

UMKC researchers should consider how their innovations align with these funding priorities when developing grant proposals and commercialization strategies. Engaging with the Missouri SBDC can help navigate these uncertainties and identify the most viable opportunities.

State, University, and Private Foundation Grants

Many private foundations and university-backed programs offer funding for early-stage research and commercialization efforts. These grants can provide critical early-stage support, particularly for technologies in the proof-of-concept phase.

Missouri Technology Corporation (MTC) provides state-level support for innovative startups through matching funds that complement federal and private investments.

- **How It Works**: MTC offers funding to startups that have secured other sources of investment, such as SBIR/STTR grants or venture capital. These matching funds help reduce the financial burden on early-stage companies and enable them to accelerate commercialization efforts.
- **Why It Matters for UMKC Startups**: University-affiliated startups can leverage MTC matching funds to expand their research, hire personnel, and advance product development. This additional funding can increase a company's ability to attract investors by demonstrating strong support from both federal and state sources.

Proof-of-Concept Funds

Proof-of-concept (POC) funds are critical resources that help early-stage startups validate their ideas and develop working prototypes before seeking larger investments. These funds provide financial support for activities such as prototype development, market validation, customer discovery, and early-stage business formation.

One such example is **Digital Sandbox KC**, a program that offers proof-of-concept funding of up to $20,000 to early-stage startups in the Kansas City region. The goal of Digital Sandbox is to reduce the financial risk associated with early-stage innovation by providing grants for critical activities that help move an idea closer to commercialization. Unlike traditional investment sources, Digital

Sandbox funding is typically non-dilutive, meaning startups do not have to give up equity in exchange for financial support.

These funds are particularly useful for university-affiliated startups, as they provide a much-needed financial bridge between research and commercialization. Entrepreneurs can use the funding to refine their business model, conduct feasibility studies, and develop minimum viable products (MVPs) that demonstrate the viability of their technology in the real world.

Missouri SBDC and UMKC Entrepreneurship Grants

The Missouri Small Business Development Center (SBDC) and UMKC's Entrepreneurship Innovation Grant provide financial support and mentorship for startups. These resources help entrepreneurs refine their business models and access additional funding sources.

Stages of Equity-Based Funding

Equity-based funding involves securing investment from external parties in exchange for partial ownership of the startup. This type of funding is crucial for scaling a business but requires careful negotiation to retain strategic control.

Startups that seek equity-based funding typically progress through several funding stages, each with unique opportunities and challenges.

Funding availability depends on a startup's stage of development. The following table outlines what investors expect at each stage:

Startup Stage	Funding Sources	Key Milestones
Idea/Research	Grants, POC Funds	Proof of concept, initial research, patent filings
Pre-Seed	Friends & Family, SBIR/STTR	Market validation, early prototype, team formation
Seed	Angel Investors, Accelerators	Minimum viable product (MVP), early customer traction, business model validation
Early Stage (Series A)	Venture Capital, Incubators	Scalable product, growing customer base, revenue generation
Growth Stage (Series B/C)	Venture Capital, Private Equity	Market expansion, strong revenue, profitability pathway
Late Stage	Private Equity, IPO	Market dominance, acquisition or IPO readiness

Below is an overview of the major funding stages and the available sources of capital for each.

Pre-Seed Funding

Pre-seed funding covers early expenses like market research and developing a minimal viable product (MVP). Most pre-seed funding comes from founders' savings or personal networks.

- **Bootstrapping**: Using personal savings or revenue generated by the business to fund operations is common at this stage. Founders maintain full control, but limited resources can slow growth.
- **Friends and Family**: This form of funding is typically the easiest to secure but requires careful consideration of legal agreements to avoid misunderstandings. Friends and family investors may not expect significant returns but can provide a quick cash infusion.

Seed Funding

Seed funding helps startups cover early-stage development costs such as product design and team hiring.

- **Angel Investors**: Wealthy individuals who invest in exchange for equity. In addition to funding, angel investors often offer valuable mentorship and industry connections.
- **Incubators and Accelerators**: Startup accelerators provide funding, mentorship, and networking opportunities to help startups refine their products and scale operations. Incubators add the element of "collision" space, helping startup entrepreneurs benefit from proximity to other growing companies.

Early-Stage Funding

Early-stage funding is critical for scaling operations and moving from a prototype to a fully commercialized product.

- **Venture Capital (VC)**: Venture capital firms invest in high-growth startups with proven business models. In exchange for equity, VCs provide large sums of money and often take an active role in shaping the company's direction. Venture capitalists are looking for scalability, a large market opportunity, and a clear exit strategy.
- **Incubators and Accelerators**: These programs not only provide funding but also mentorship, office space, and networking opportunities to help startups scale quickly. Accelerators often take equity in exchange for services and may culminate in a demo day where startups pitch to investors.

Growth Stage Funding

Once a startup achieves product-market fit and is ready to scale further, growth-stage funding provides the capital necessary to expand.

- **Series A, B, and C Funding**: As a startup grows, it may go through multiple rounds of venture funding, with each round increasing in size. Series A focuses on scaling operations, while Series B and C target geographic expansion and product diversification. For instance, Series A funding typically ranges from $2 million to $15 million. Later rounds, such as Series B or C, can involve tens of millions of dollars.

Late-Stage Funding

For mature startups, late-stage funding helps them expand globally, acquire other companies, or prepare for an IPO.

- **Private Equity**: Private equity funds invest in more established startups, often preparing them for acquisitions or public offerings. These funds typically seek higher returns and a clear exit strategy, either through an acquisition or IPO.
- **Initial Public Offering (IPO)**: When a company goes public, it offers shares to the general market. This is a common exit strategy for startups at the peak of their growth but carries significant risks, including market volatility.

Alternative Funding Sources

For startups that may not qualify for venture capital or prefer to explore alternative funding options, there are several other sources of capital.

- **Crowdfunding**: Platforms like Kickstarter and Indiegogo allow startups to raise capital by pitching their ideas to the public. This method is particularly useful for consumer-facing products and helps test market demand before full-scale production. Equity crowdfunding platforms such as SeedInvest or Crowdcube allow startups to raise funds in exchange for shares. The most successful crowdfunded businesses have a few things in common:
    - **Don't start without 75%+ committed** – Crowdfunding campaigns rely on the excitement generated around an innovation, and products that show significant jumps in the first 48 hours of the campaign rise to the top of the promotion list.
    - **Set a realistic goal, blow it away** – Realize that with most platforms, you won't receive the funds unless you have met your fundraising goal. So set a realistic goal and hit it as quickly as possible with pre-loaded interest. If there's enough interest, it may grow to a multiple of your original goal. Know what you need to close versus what you'd like to close.
    - **Promote Aggressively**: Use social media, press releases, and influencer partnerships to extend your campaign's reach. Regular updates keep backers engaged and build trust, increasing the likelihood of additional support and future funding opportunities. Encourage additional funding by offering new incentives when initial goals are met.

- **Loans**: Startups can apply for small business loans through the Small Business Administration (SBA). SBA loans provide low-interest rates and favorable terms for qualifying startups. Other loan options include credit card loans or short-term loans to cover immediate cash flow needs. The challenge is that banks only loan money if you can demonstrate the ability to pay the loan back; for many pre-profit startups, bank loans are not a viable avenue of funding.

- **Pitch Competitions**: Pitch competitions provide an excellent opportunity for startups to secure funding while gaining visibility and feedback from investors. UMKC and other organizations host competitions where entrepreneurs pitch their ideas to panels of judges for a chance to win funding and mentorship. Programs such as the Regnier Venture Creation Challenge and national competitions like Rice Business Plan Competition offer cash prizes, networking opportunities, and investor connections to promising startups.

UMKC-Specific Resources

UMKC offers several tailored resources to help faculty and students navigate the funding landscape.

Guiding Question:
What kind of funding or support would help move my work closer to real-world use, and what milestone would that funding need to accomplish?

- **Office of Commercialization**: UMKC's Office of Commercialization (OC) helps university researchers and entrepreneurs develop business plans, connect with investors, and secure funding. The OC works in tandem with UMKC's Office of Tech Transfer (OTT) to provide expert guidance on securing intellectual property rights and finding commercialization partners.
- **Technology Venture Studio**: UMKC's Technology Venture Studio (TVS) is part of the UMKC Innovation Center. With programs like Digital Sandbox ($20,000 in proof-of-concept funding), Whiteboard2Boardroom, and the Office of Innovation Impact, TVS provides expert guidance, funding, and access to a network of entrepreneurs and investors.
- **Missouri SBDC:** The Missouri SBDC provides workshops, business counseling, and funding guidance tailored to university entrepreneurs.
- **Entrepreneurship Innovation Grant:** The Entrepreneurship Innovation Grant (EIG) is designed to support innovative university-affiliated startups by offering financial assistance for prototype development, market validation, and business planning. This grant aims to bridge the gap between research and commercialization by providing seed capital to promising entrepreneurial projects. Unlike traditional grant programs focused solely on academic research, this grant emphasizes the commercial potential of innovations, helping faculty and students transition their ideas into viable business opportunities.
- **Regnier Venture Creation Challenge:** UMKC's Bloch School of Business hosts the annual Regnier Venture Creation Challenge, offering cash prizes and funding for top startup ideas. There are tracks for UMKC students and community entrepreneurs.

Strategies for Securing Funding

Successfully securing funding involves more than just identifying the right sources. University startups must develop strong strategies to make their businesses attractive to investors.

- **Build a Strong Team**: Investors look for well-rounded teams with both technical and business expertise. Founders should focus on building teams with a strong "founder-market fit," meaning team members are deeply familiar with the problems they aim to solve. Investors look for well-balanced teams with a mix of technical expertise, business acumen, and execution capability. The ideal startup team includes:
    - **Technical Lead (University Researcher/Inventor)**: Provides deep expertise in the technology, innovation, and research aspects.

- **Business Lead (CEO or Entrepreneurial Co-Founder):** Manages business strategy, fundraising, and market expansion.
- **Product/Operations Lead:** Ensures the product is developed efficiently and delivered to market.
- **Sales & Marketing Lead:** Drives customer acquisition and revenue growth.

Faculty members and graduate students often serve as the technical lead, developing and refining the innovation. However, they may need to partner with experienced business professionals to manage commercialization and investor relations effectively.

- **Develop a Clear Exit Strategy**: Investors, particularly venture capitalists and angel investors, want to know how and when they will see a return on their investment. A clear path to acquisition, IPO, or licensing deal can be a strong selling point.

- **Leverage Grants and University Resources**: Securing non-dilutive funding such as SBIR or STTR grants is highly attractive to venture capitalists, as it means the startup requires less external funding and retains more equity. UMKC's support in grant writing and the availability of internal university funding sources can be invaluable in this respect.

Conclusion

Funding can play an important role in helping research move toward real-world use, but the most useful funding is tied to a specific purpose. In some cases, that purpose is technical validation. In others, it is market discovery, prototype development, team building, partnership readiness, or startup formation. The goal is not simply to secure money. The goal is to reduce uncertainty, reach the next meaningful milestone, and strengthen the pathway through which your work could create impact.

At UMKC, faculty and graduate students do not need to have a company, a polished pitch, or a complete plan in order to begin that conversation. If your work may have potential beyond the university, the Office of Commercialization can help you think through what kind of support makes sense, what stage you are in, and what next step is most worth pursuing.

Resources & Contacts

UMKC Office of Commercialization
Director: Jeff Blackwood, MBA
📍 Website: www.umkcinnovates.com/tech-commercialization
📧 Email: techcomm@umkc.edu
📞 Phone: 816-235-6373